Backlog

Applied manufactures systems to meet demand represented by order backlog and customer commitments. Backlog consists of: (1) orders for which written authorizations have been accepted and assigned shipment dates are within the next 12 months, or shipment has occurred but revenue has not been recognized; and (2) contractual service revenue and maintenance fees to be earned within the next 12 months.

Backlog by reportable segment as of October 27, 2013 and October 28, 2012 was as follows:

	2013		2012	
	(In millions, except percentages)			
Silicon Systems Group	$ 1,295	55%	$ 705	44%
Applied Global Services	591	25%	580	36%
Display	361	15%	206	13%
Energy and Environmental Solutions	125	5%	115	7%
Total	$ 2,372	100%	$ 1,606	100%

Applied's backlog on any particular date is not necessarily indicative of actual sales for any future periods, due to the potential for customer changes in delivery schedules or cancellation of orders. Customers may delay delivery of products or cancel orders prior to shipment, subject to possible cancellation penalties. Delays in delivery schedules and/or a reduction of backlog during any particular period could have a material adverse effect on Applied's business and results of operations.

Manufacturing, Raw Materials and Supplies

Applied's manufacturing activities consist primarily of assembly, test and integration of various proprietary and commercial parts, components and subassemblies (collectively, parts) that are used to manufacture systems. Applied has implemented a distributed manufacturing model under which manufacturing and supply chain activities are conducted in various countries, including the United States, Europe, Israel, Singapore, Taiwan, and other countries in Asia, and assembly of some systems is completed at customer sites. Applied uses numerous vendors, including contract manufacturers, to supply parts and assembly services for the manufacture and support of its products. Although Applied makes reasonable efforts to assure that parts are available from multiple qualified suppliers, this is not always possible. Accordingly, some key parts may be obtained from only a single supplier or a limited group of suppliers. Applied seeks to reduce costs and to lower the risks of manufacturing and service interruptions by: (1) selecting and qualifying alternate suppliers for key parts; (2) monitoring the financial condition of key suppliers; (3) maintaining appropriate inventories of key parts; (4) qualifying new parts on a timely basis; and (5) locating certain manufacturing operations in close proximity to suppliers and customers.

Research, Development and Engineering

Applied's long-term growth strategy requires continued development of new products. The Company's significant investment in research, development and engineering (RD&E) has generally enabled it to deliver new products and technologies before the emergence of strong demand, thus allowing customers to incorporate these products into their manufacturing plans at an early stage in the technology selection cycle. Applied works closely with its global customers to design systems and processes that meet their planned technical and production requirements. Product development and engineering organizations are located primarily in the United States, as well as in Europe, Israel, Taiwan, and China. In addition, Applied outsources certain RD&E activities, some of which are performed outside the United States, primarily in India. Process support and customer demonstration laboratories are located in the United States, China, Taiwan, Europe, and Israel.

Applied's investments in RD&E for product development and engineering programs to create or improve products and technologies over the last three years were as follows: $1.3 billion (18 percent of net sales) in fiscal 2013, $1.2 billion (14 percent of net sales) in fiscal 2012, and $1.1 billion (11 percent of net sales) in fiscal 2011. Applied has spent an average of 14 percent of net sales in RD&E over the last five years. In addition to RD&E for specific product technologies, Applied maintains ongoing programs for automation control systems, materials research, and environmental control that are applicable to its products.